Android Forensic Analysis Using Open-Source Tools

Project Proposal

Umesh Kalupathirannehelage – 300389749 (Team Lead)

Sahan Pattinikuttige Nonis – 300389470

Group 7

Mobile Cybersecurity

CSIS4440 - 003

Table of Contents

1. Tools and Apps Overview

1.1 Student: Umesh Kalupathirannehelage

Tools: ALEAPP (Android Logs, Events & Protobuf Parser), MobSF (Mobile Security Framework), LiME (Linux Memory Extractor) – optional if time permits.

Apps: Dropbox; Strava.

Novelty & Value Addition:

• Focus on cloud storage artifacts (Dropbox) and GPS fitness traces (Strava) to cover diverse evidence types.
• Use MobSF first to identify permissions, APIs, and storage paths; then use ALEAPP to parse actual artifacts and build timelines.
• Optionally use LiME to capture volatile evidence (in-memory tokens, decrypted fragments) that disk-only analysis may miss.
• Correlate timestamps across apps to reconstruct user activity with higher confidence.

1.2 Student: Sahan Pattinikuttige Nonis

Tools: JADX, Drozer, Androguard – optional if time permits.

Apps: Trust Wallet; Spotify.

Novelty & Value Addition:

• Combine reverse engineering (JADX/Androguard) with dynamic assessment (Drozer) to map code behavior to device artifacts.
• Analyze two distinct surfaces: a crypto wallet (Trust Wallet) and a media streaming client (Spotify).
• Identify potential weak storage points or exported components that may influence forensic artifact locations.

2. Proposed Integrated Component

We will produce a Combined Forensic Dashboard aggregating both students' findings into a single view. The dashboard will be delivered as a CSV/Excel or simple Python-generated report summarizing:

• Extracted artifact locations, database paths, and cache directories
• Key timestamps (creation, modification, access)
• App-specific indicators (e.g., tokens, GPX/GPX-derived points, sync events)
• Cross-app comparisons and correlations (e.g., Strava GPS activity around the same time as Dropbox sync activity.

3. AI Use

Table of AI Tools and Specific Use:

AI Tool Name	Version, Account Type	Specific feature for which the AI tool was used
ChatGPT	GPT-5 (Plus)	Used to research the capabilities of forensic tools (ALEAPP, MobSF, Drozer, JADX), understand how they can be applied to Android applications, and explore possible integration approaches for combining both students' results into a single dashboard.

While AI was used to research tool capabilities and brainstorm integration ideas, all hands-on forensic work will be performed manually by us. This includes:

- Setting up and operating a rooted Android emulator to generate real forensic artifacts
- Executing ALEAPP, MobSF, Drozer, and JADX personally, without AI automation
- Performing actual data extraction, analysis, and verification from the selected apps (Dropbox, Strava, Trust Wallet, Spotify)
- Manually creating the Combined Forensic Dashboard based on real outputs and logs
- Writing our own findings, screenshots, interpretations, and final analysis

AI will not perform any technical execution, forensic commands, or analysis of evidence.

Appendix: Full prompt history will be included in the appendix of the final report.

4. Project Contract

Meeting Frequency	Weekly, with ad-hoc syncs as needed.
Communication	Discord/WhatsApp and in-person coordination after class.
Responsibilities	Each student completes their own tool/app analysis; both collaborate on the Combined Dashboard and final report.
Professional Conduct	Respectful, timely updates, equal workload, shared decisions.

Missed Work	Member not meeting a deadline must notify partner within 24 hours and propose a catch-up plan.
Submissions	Team Lead (Umesh) submits to Blackboard and the GitHub repository as required.

By submitting this project to Blackboard and uploading it to the official GitHub repository, both team members confirm that they have read, understood, and agreed to the terms outlined in this Project Contract. We jointly accept responsibility for our individual contributions, collaborative work, communication commitments, and academic integrity throughout the duration of the project.

5. Work Log Table

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Date	Student	Work Summary	Hours
15th Sep 2025	Both	Group formed, agreed on roles.	0.5
6th Oct 2025	Umesh	Researched about tools and apps which can be used for the project.	1
6th Oct 2025	Sahan	Researched about tools and apps.	1
7th Oct 2025	Umesh	Drafted an email to professor to finalize the tools and applications which will be used.	0.5
9th Oct 2025	Both	Met with professor to discuss scope and confirm the usage of tools and apps.	0.5
21st Oct 2025	Umesh	Tool research and setup: ALEAPP, MobSF, drafted	1

		extraction workflow for Dropbox & Strava.	
22nd Oct 2025	Sahan	Tool research and setup: Drozer, JADX, drafted assessment plan for Trust Wallet & Spotify.	1

6. References

6.1 Tools

Brignoni, A. (2020). *ALEAPP – Android Logs, Events, and Protobuf Parser* [Computer software]. GitHub. https://github.com/abrignoni/ALEAPP

Mobile Security Framework (MobSF). (2024). *Mobile Security Framework (MobSF)* [Computer software]. GitHub. https://github.com/MobSF/Mobile-Security-Framework-MobSF

FSecure Labs. (2015). *Drozer* [Computer software]. GitHub. https://github.com/FSecureLABS/drozer

Skylot. (2024). *JADX – Dex to Java decompiler* [Computer software]. GitHub. https://github.com/skylot/jadx

Androguard Project. (2024). *Androguard – Reverse engineering and malware analysis of Android applications* [Computer software]. GitHub. https://github.com/androguard/androguard

504ensics Labs. (2014). *LiME – Linux Memory Extractor* [Computer software]. GitHub. https://github.com/504ensicsLabs/LiME

6.2 Apps

Dropbox. (2024). *Dropbox (Android app)* [Mobile application]. Google Play Store. https://play.google.com/store/apps/details?id=com.dropbox.android

Strava. (2024). *Strava (Android app)* [Mobile application]. Google Play Store. https://play.google.com/store/apps/details?id=com.strava

Trust Wallet. (2024). *Trust Wallet (Android app)* [Mobile application]. Google Play Store. https://play.google.com/store/apps/details?id=com.wallet.crypto.trustapp

Spotify. (2024). *Spotify (Android app)* [Mobile application]. Google Play Store. https://play.google.com/store/apps/details?id=com.spotify.music

Telegram FZ-LLC. (2024). *Telegram (Android app)* [Mobile application]. Google Play Store. https://play.google.com/store/apps/details?id=org.telegram.messenger

7. Appendix

Prompts used on ChatGPT (Umesh)

“What are the commonly used open-source and commercial tools available for Android mobile forensic analysis and what types of data they can extract. Exclude the following list.

db
adb backup extractor
Magnet Acquire
Autopsy
Carbon
HBE
Helium Android
Nandump
FTK Imager
Android Backup Extractor
DCode
DB Browser Lite
JsonViewer
vtuploader
yara
hasher
nmap - network mapper”

“Explain the capabilities of ALEAPP & MobSF in mobile forensic investigations”

“Is it realistic to perform Android data extraction and using a rooted emulator? Confirm whether tools like ALEAPP and MobSF will produce valid results.”

“Do apps like Dropbox, Strava, generate useful forensic artifacts? I want to confirm they can be analysd with ALEAPP and MobSF.”

“Suggest possible ways for two students to integrate android forensic findings into one final output. I’m using ALEAPP/MobSF and other teammate is doing Drozer/JADX. I want ideas to create a integrated component.”